

1. Suppose $P(A) = 0.5$, $P(B) = 0.4$, and $P(A \cap B) = 0.2$. What is $P(A \cup B)$?
 - A. 0.5
 - B. 0.7
 - C. 0.8
 - D. 0.9

2. Suppose $P(A) = 0.6$, $P(B) = 0.3$, and $P(A \cap B) = 0.18$. What is $P(A \mid B)$?
 - A. 0.18
 - B. 0.30
 - C. 0.60
 - D. 0.90

3. Suppose $P(A) = 0.45$, $P(B) = 0.35$, and $P(A \cap B) = 0.10$. What is the probability that exactly one of A and B occurs?
 - A. 0.25
 - B. 0.45
 - C. 0.60
 - D. 0.80

4. Suppose $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \cap B) = 0.2$. Which statement is correct?
 - A. A and B are independent
 - B. A and B are mutually exclusive
 - C. A and B cannot both occur
 - D. $P(A \mid B) = 0.2$

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5. A fair six-sided die is rolled twice. What is the probability that the sum is even?
- A. $1/4$
 - B. $1/3$
 - C. $1/2$
 - D. $3/4$
6. A fair six-sided die is rolled twice. What is the probability that the maximum number rolled is at least 5?
- A. $1/9$
 - B. $1/3$
 - C. $5/9$
 - D. $3/4$
7. Let $X \sim \text{Binomial}(20, 0.4)$. What is $E[X]$?
- A. 4
 - B. 6
 - C. 8
 - D. 10
8. Let $X \sim \text{Binomial}(20, 0.4)$. What is $\text{Var}(X)$?
- A. 2.4
 - B. 4.8
 - C. 8
 - D. 12

9. Let $X \sim \text{Binomial}(5, 0.2)$. What is $P(X = 0)$?

- A. $(0.2)^5$
- B. $5(0.2)(0.8)^4$
- C. $(0.8)^5$
- D. $1 - (0.8)^5$

10. Let $X \sim \text{Poisson}(3)$. What is $E[X^2]$?

- A. 3
- B. 6
- C. 9
- D. 12

11. Let $X \sim \text{Poisson}(2)$. What is $P(X \leq 1)$?

- A. e^{-2}
- B. $2e^{-2}$
- C. $3e^{-2}$
- D. $1 - 3e^{-2}$

12. Let X have pmf

$$p(x) = c(x + 1), \quad x = 0, 1, 2.$$

What is c ?

- A. $1/6$
 - B. $1/3$
 - C. $1/2$
 - D. 1
13. For the same random variable, what is $P(X \geq 1)$?
- A. $1/6$
 - B. $1/3$
 - C. $2/3$
 - D. $5/6$
14. For the same random variable, what is $E[X]$?
- A. $1/2$
 - B. 1
 - C. $4/3$
 - D. 2
15. For the same random variable, what is $\text{Var}(X)$?
- A. $1/9$
 - B. $5/9$
 - C. $4/3$
 - D. $7/3$

16. Let X have pdf

$$f(x) = cx, \quad 0 \leq x \leq 4.$$

What is c ?

- A. $1/8$
- B. $1/4$
- C. $1/2$
- D. 1

17. For the same random variable, what is $P(X \leq 2)$?

- A. $1/4$
- B. $1/2$
- C. $3/4$
- D. 1

18. For the same random variable, what is $E[X]$?

- A. 1
- B. 2
- C. $8/3$
- D. 4

19. Let X have cdf

$$F(x) = \frac{(x-1)^2}{4}, \quad 1 \leq x \leq 3.$$

What is $P(X > 2)$?

- A. $1/4$
- B. $1/2$
- C. $3/4$
- D. 1

20. Suppose $X \sim N(40, 16)$. What is $P(X < 36)$ approximately?

- A. 0.023
- B. 0.159
- C. 0.500
- D. 0.841

21. Suppose $X \sim N(40, 16)$. What is $P(32 < X < 48)$ approximately?

- A. 0.341
- B. 0.683
- C. 0.841
- D. 0.954

22. Suppose $X \sim N(100, 25)$. Approximately what value of x satisfies

$$P(X \leq x) = 0.8413?$$

- A. 95
- B. 100
- C. 105
- D. 110

23. Suppose X and Y have joint pmf

	$Y = 0$	$Y = 1$
$X = 0$	0.2	0.3
$X = 1$	0.1	0.4

What is $P(X = 1)$?

- A. 0.1
- B. 0.4
- C. 0.5
- D. 0.7

24. Using the same joint pmf, what is $P(Y = 1 \mid X = 1)$?

- A. 0.2
- B. 0.4
- C. 0.6
- D. 0.8

25. Using the same joint pmf, what is $\text{Cov}(X, Y)$?

- A. 0
- B. 0.05
- C. 0.10
- D. 0.25

26. Suppose $X_1, \dots, X_{100}$ are i.i.d. with mean $\mu = 20$ and variance $\sigma^2 = 25$. Using the CLT, what is

$$P(\bar{X} > 21)$$

approximately?

- A. 0.023
- B. 0.159
- C. 0.500
- D. 0.977

27. Suppose $X_1, \dots, X_{64}$ are i.i.d. with mean $\mu = 10$ and variance $\sigma^2 = 16$. Using the CLT, what is

$$P(\bar{X} < 9.5)$$

approximately?

- A. 0.023
- B. 0.159
- C. 0.500
- D. 0.841

28. Suppose X has mean 50 and variance 16. Using Chebyshev's inequality, what upper bound can we place on

$$P(|X - 50| \geq 8)?$$

- A. 0.0625
- B. 0.125
- C. 0.25
- D. 0.50

29. Suppose X has mean 12 and variance 9. Using Chebyshev's inequality, what lower bound can we place on

$$P(6 < X < 18)?$$

- A. 0.50
- B. 0.75
- C. 0.875
- D. 0.95

30. Suppose X and Y are random variables with

$$\text{Var}(X) = 9, \quad \text{Var}(Y) = 16,$$

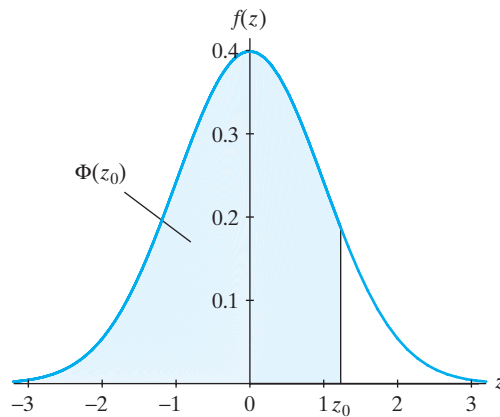
and correlation

$$\rho_{X,Y} = 0.5.$$

What is

$$\text{Var}(2X - 3Y)?$$

- A. 84
- B. 108
- C. 132
- D. 156

Table Va The Standard Normal Distribution Function

$$P(Z \leq z) = \Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-w^2/2} dw$$

$$\Phi(-z) = 1 - \Phi(z)$$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7703	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
α	0.400	0.300	0.200	0.100	0.050	0.025	0.020	0.010	0.005	0.001
z_α	0.253	0.524	0.842	1.282	1.645	1.960	2.054	2.326	2.576	3.090
$z_{\alpha/2}$	0.842	1.036	1.282	1.645	1.960	2.240	2.326	2.576	2.807	3.291